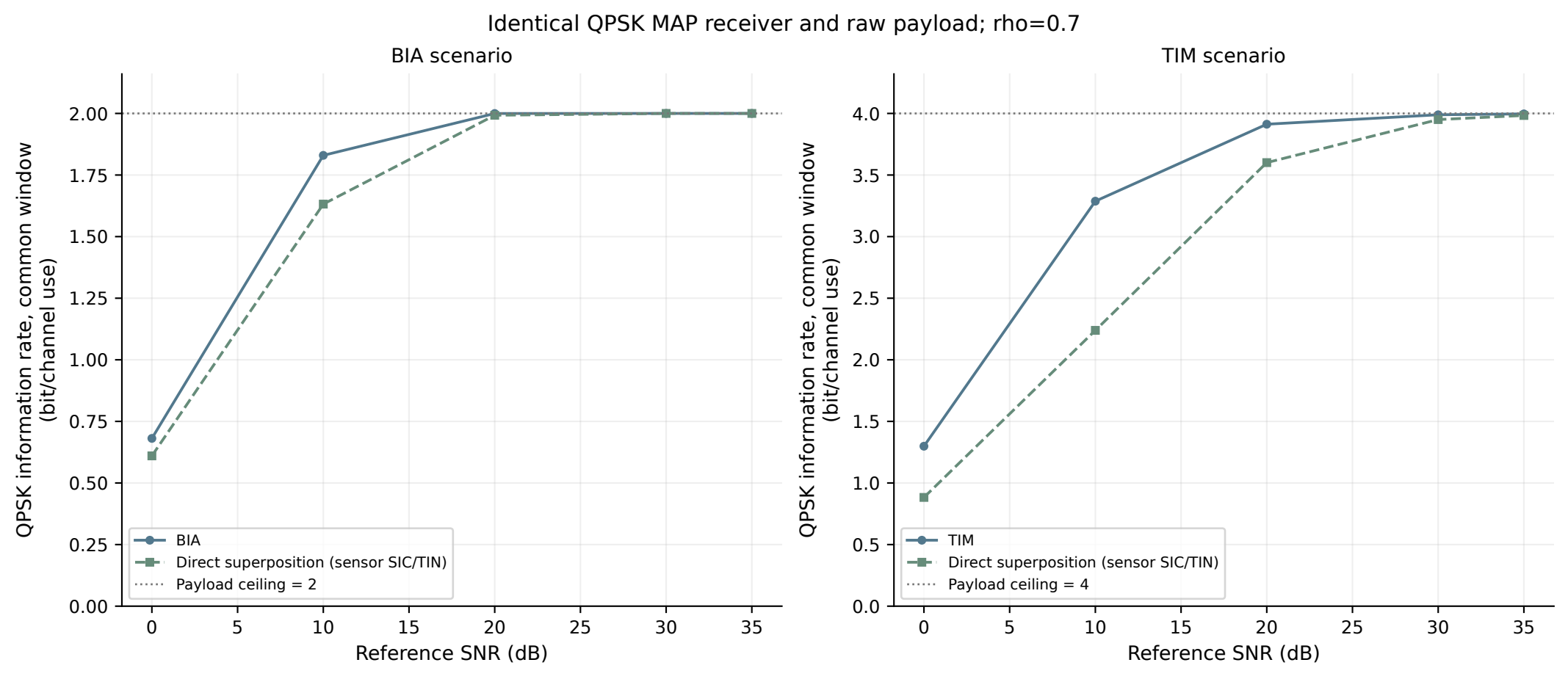
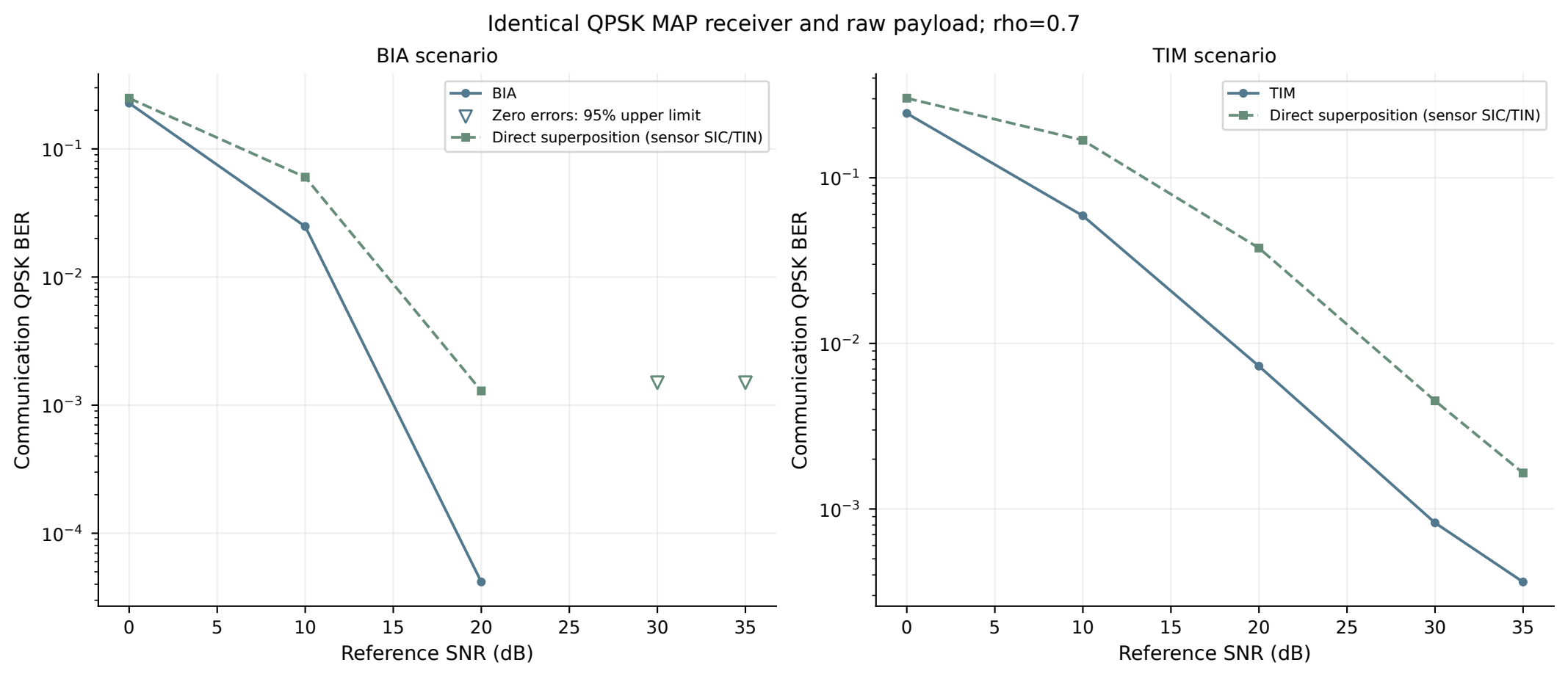
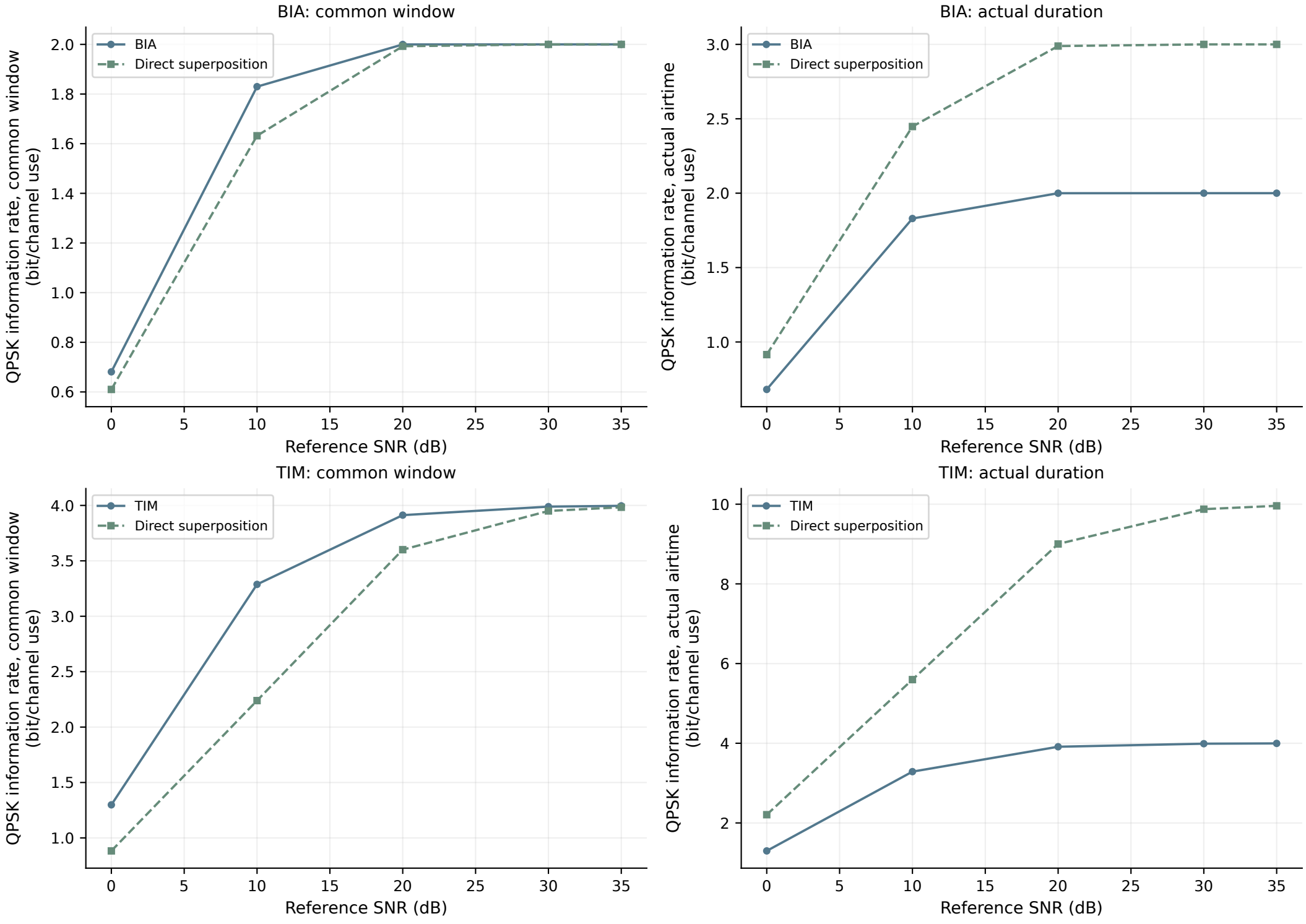
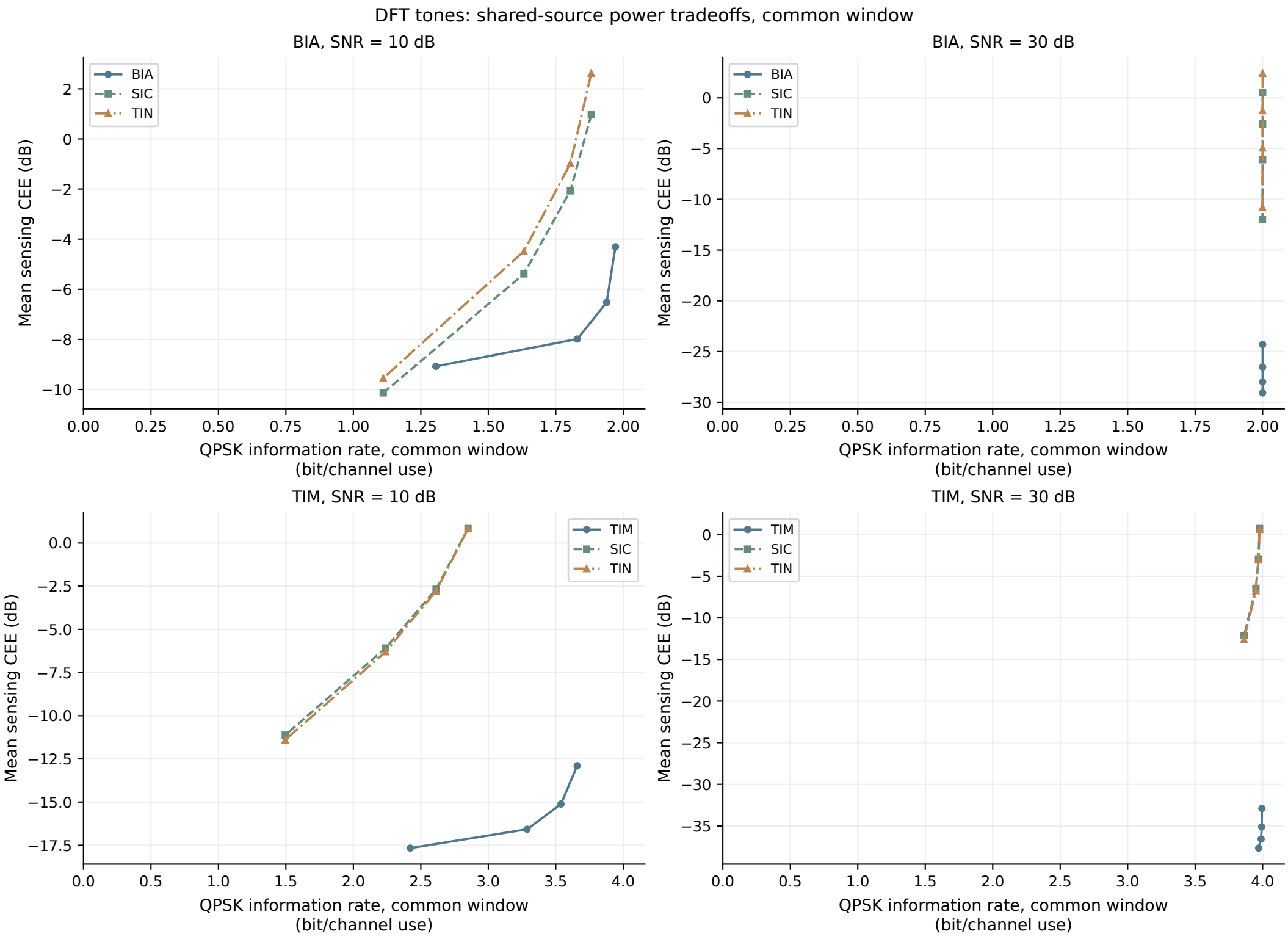
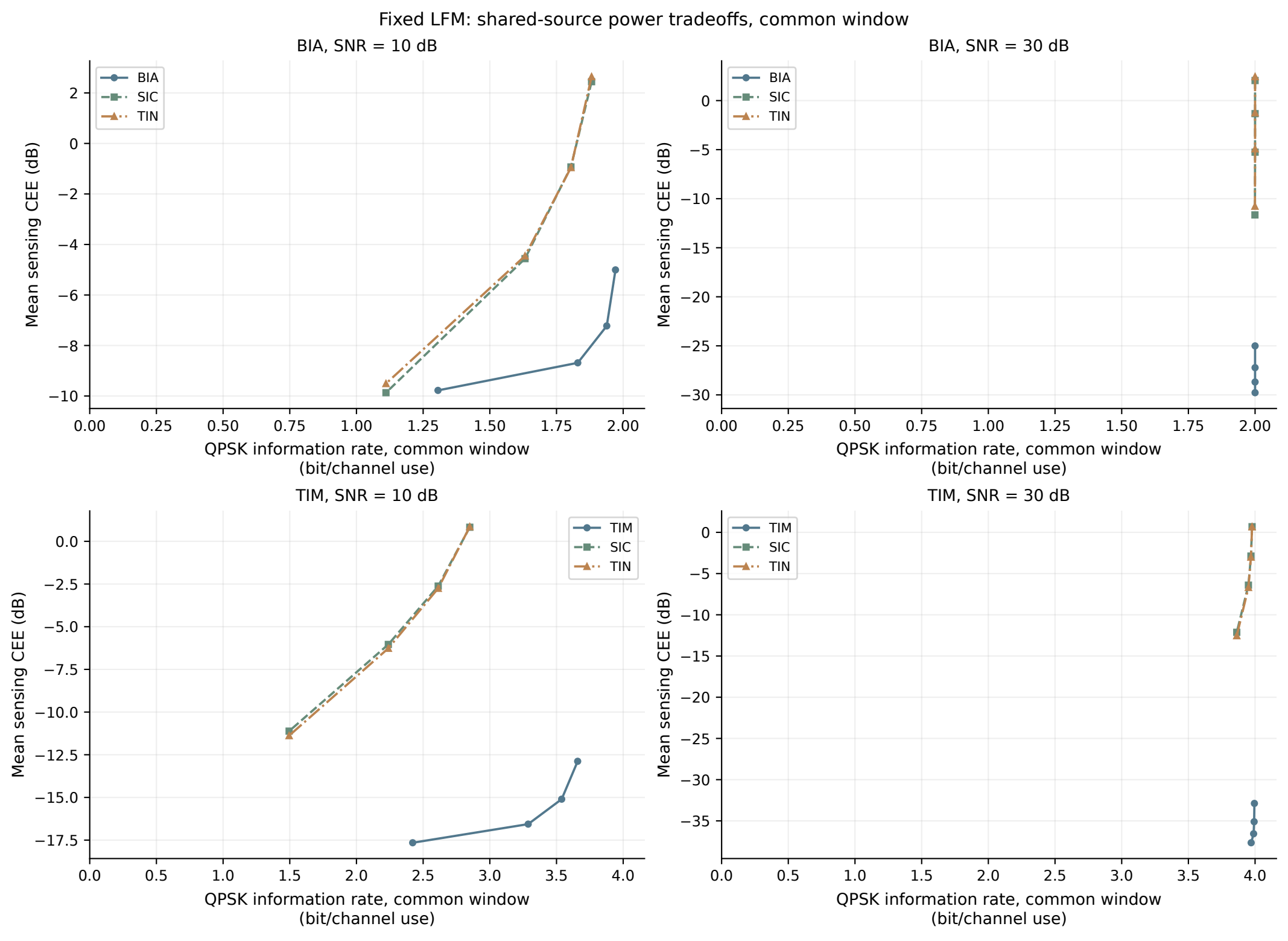
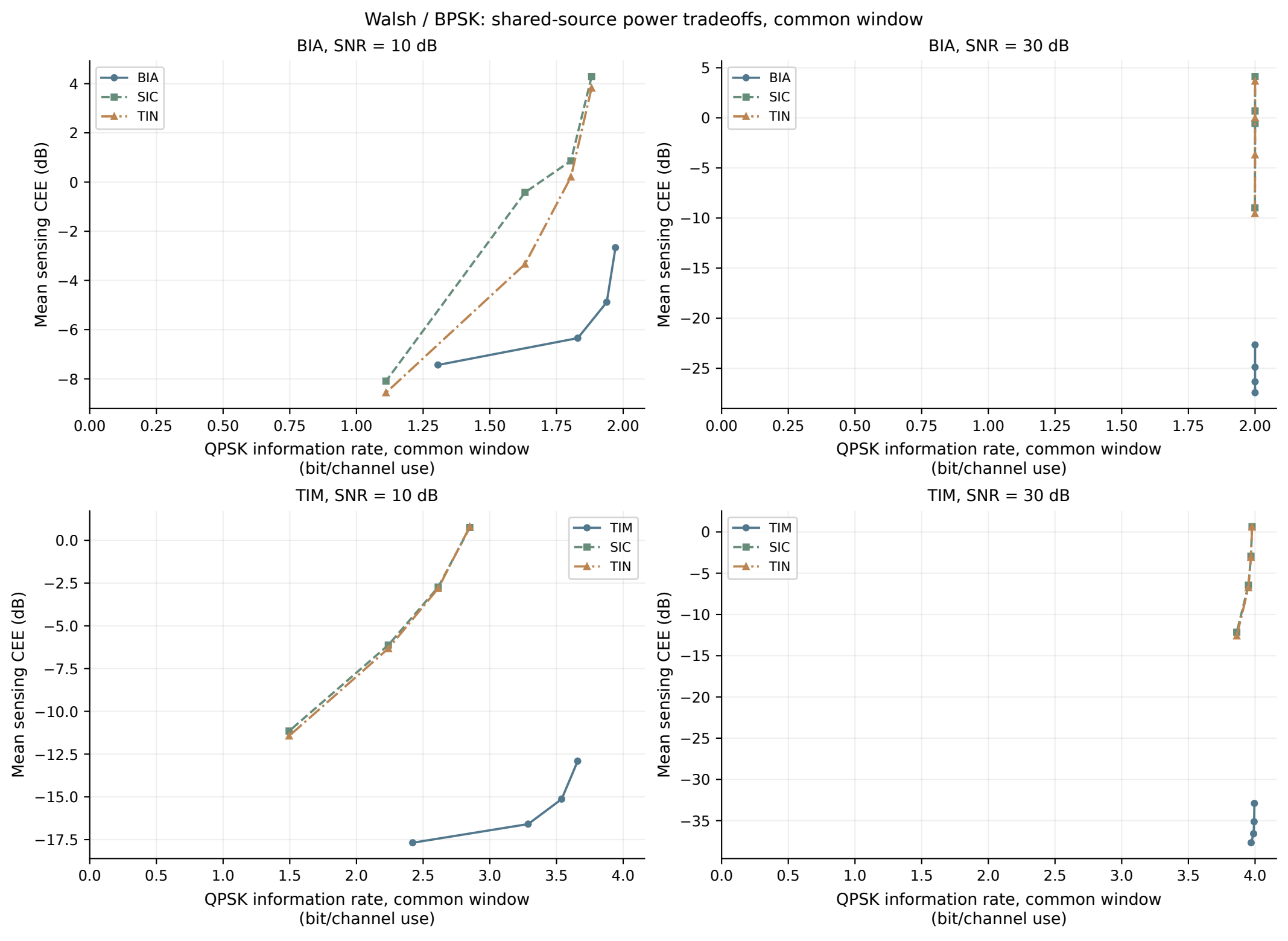

Identical QPSK MAP receiver and raw payload; rho=0.7
BIA scenario
TIM scenario
QPSK information rate, common window (bit/channel use)
Reference SNR (dB)
BIA
Direct superposition (sensor SIC/TIN)
Payload ceiling = 2
TIM
Payload ceiling = 4
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
5
10
15
20
25
30
35

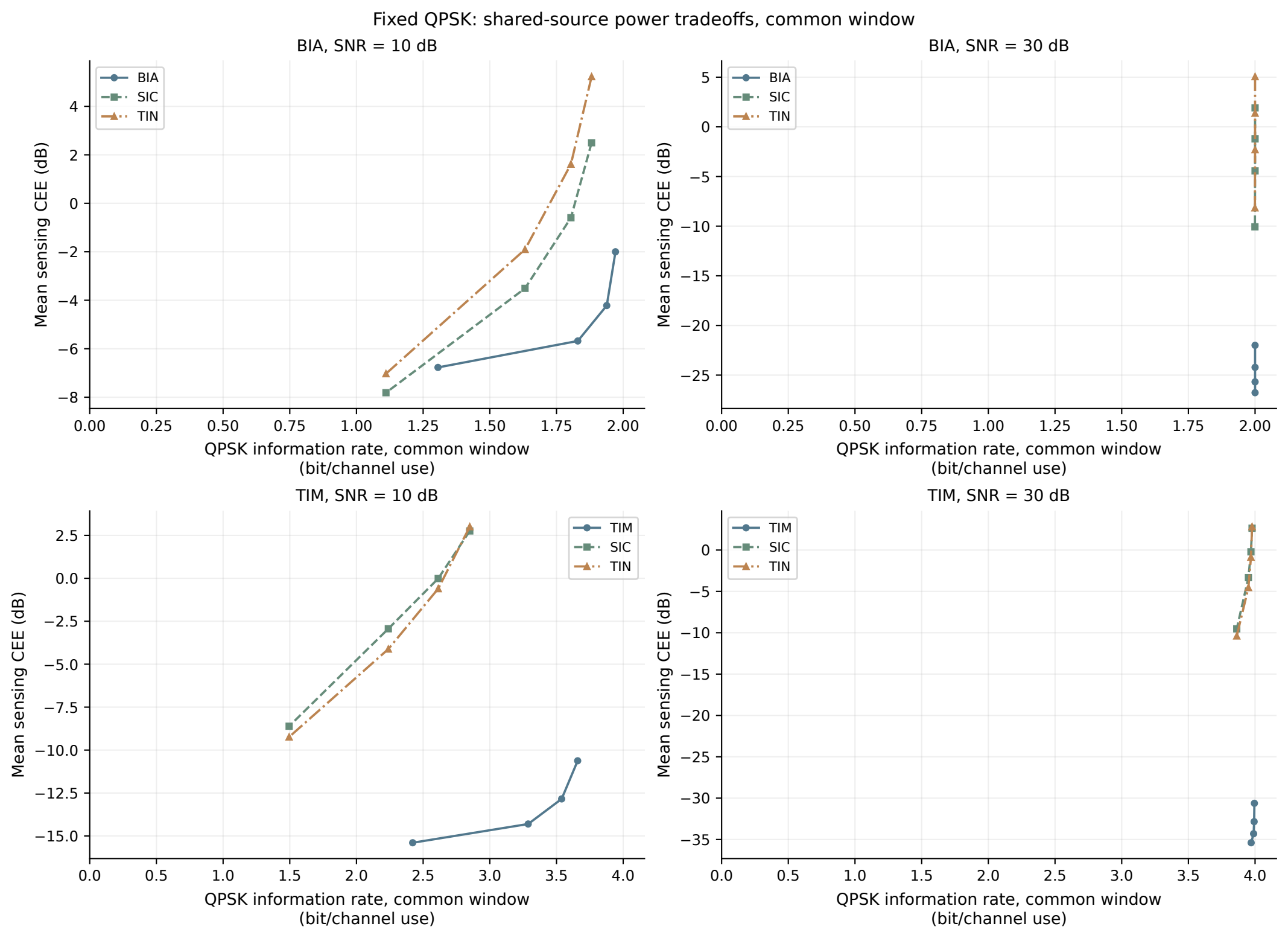

Identical QPSK MAP receiver and raw payload; rho=0.7
BIA scenario
TIM scenario
BIA
Zero errors: 95% upper limit
Direct superposition (sensor SIC/TIN)
TIM
Direct superposition (sensor SIC/TIN)
Communication QPSK BER
Reference SNR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
5
10
15
20
25
30
35

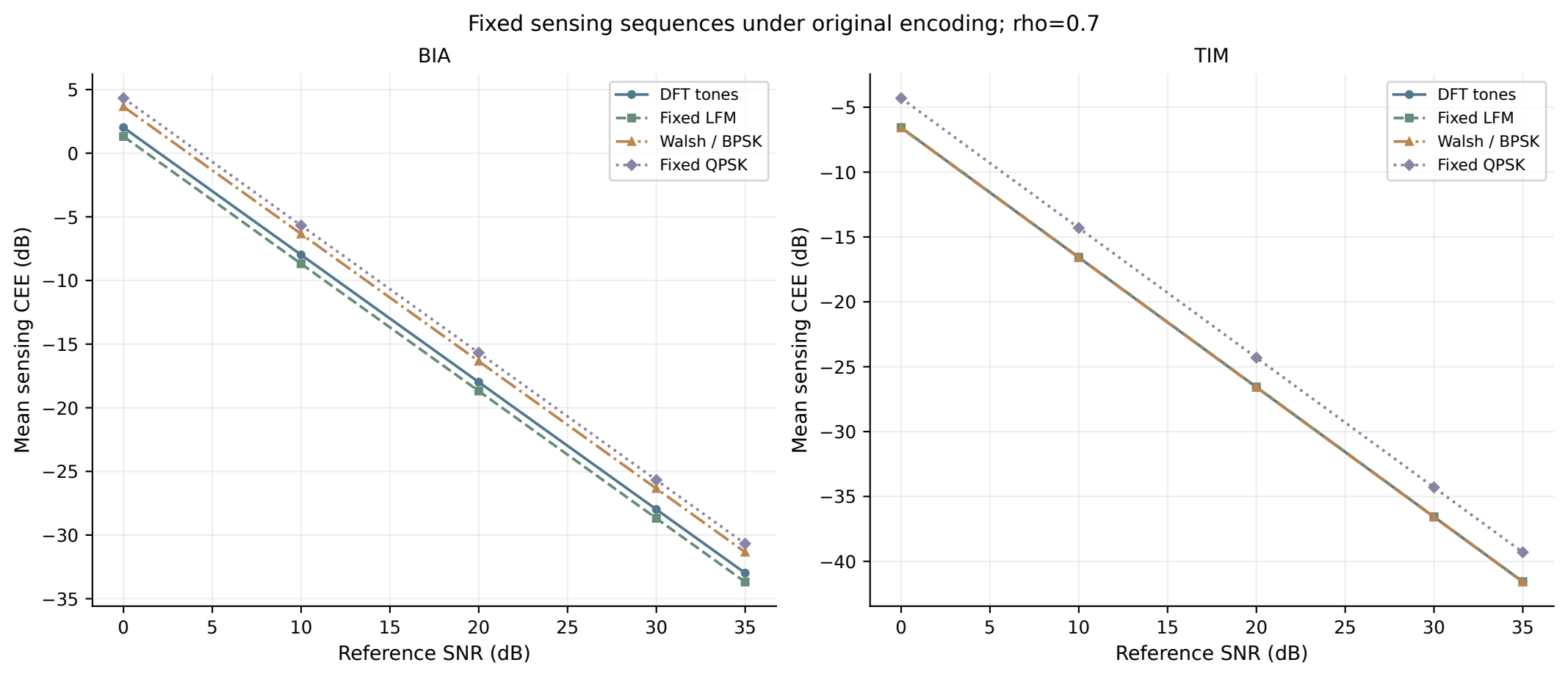

Rate normalization: common deadline versus actual airtime
BIA: common window
BIA: actual duration
TIM: common window
TIM: actual duration
BIA
Direct superposition
TIM
QPSK information rate, common window (bit/channel use)
QPSK information rate, actual airtime (bit/channel use)
Reference SNR (dB)

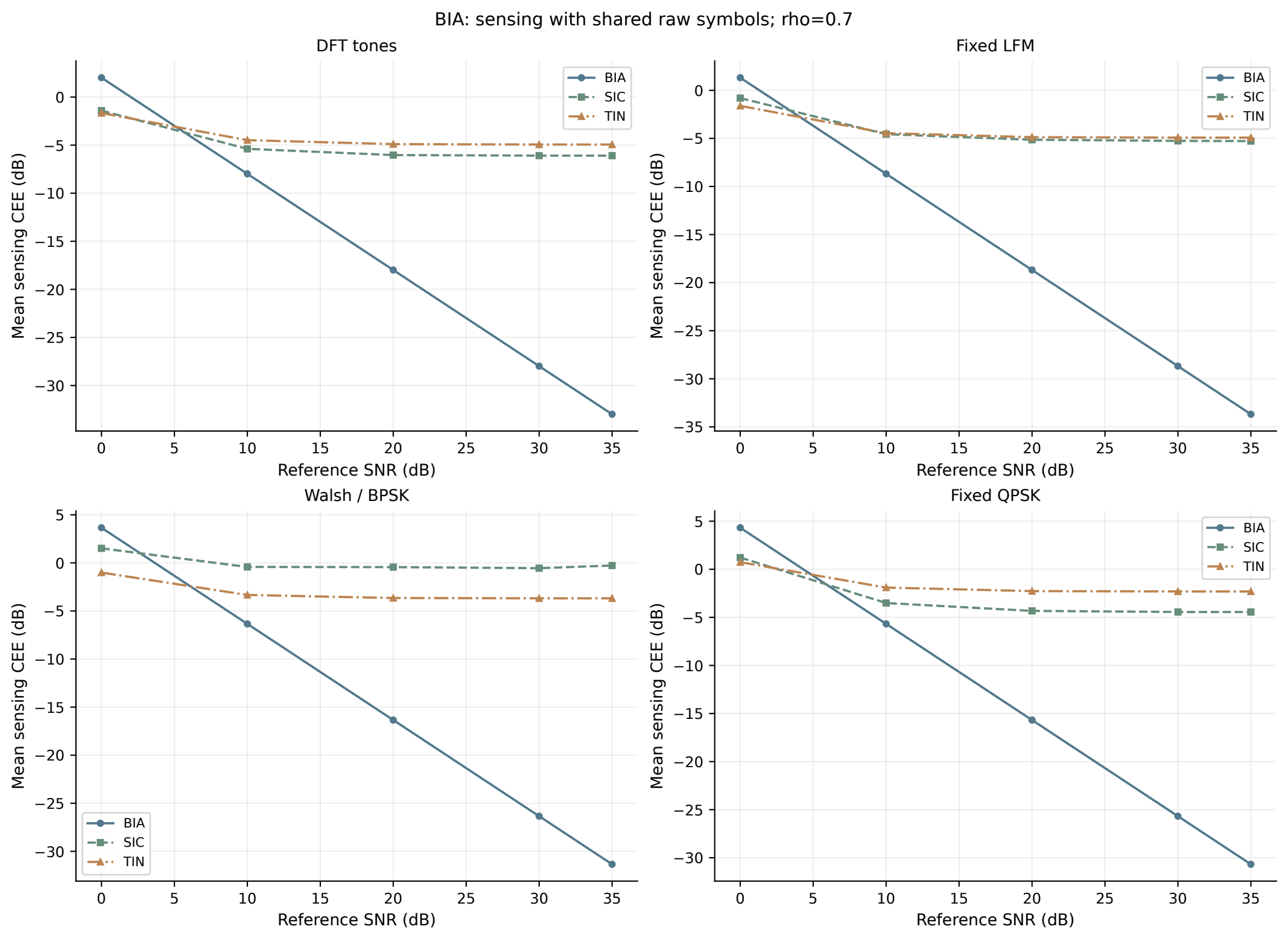

DFT tones: shared-source power tradeoffs, common window
BIA, SNR = 10 dB
BIA, SNR = 30 dB
TIM, SNR = 10 dB
TIM, SNR = 30 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, common window
(bit/channel use)

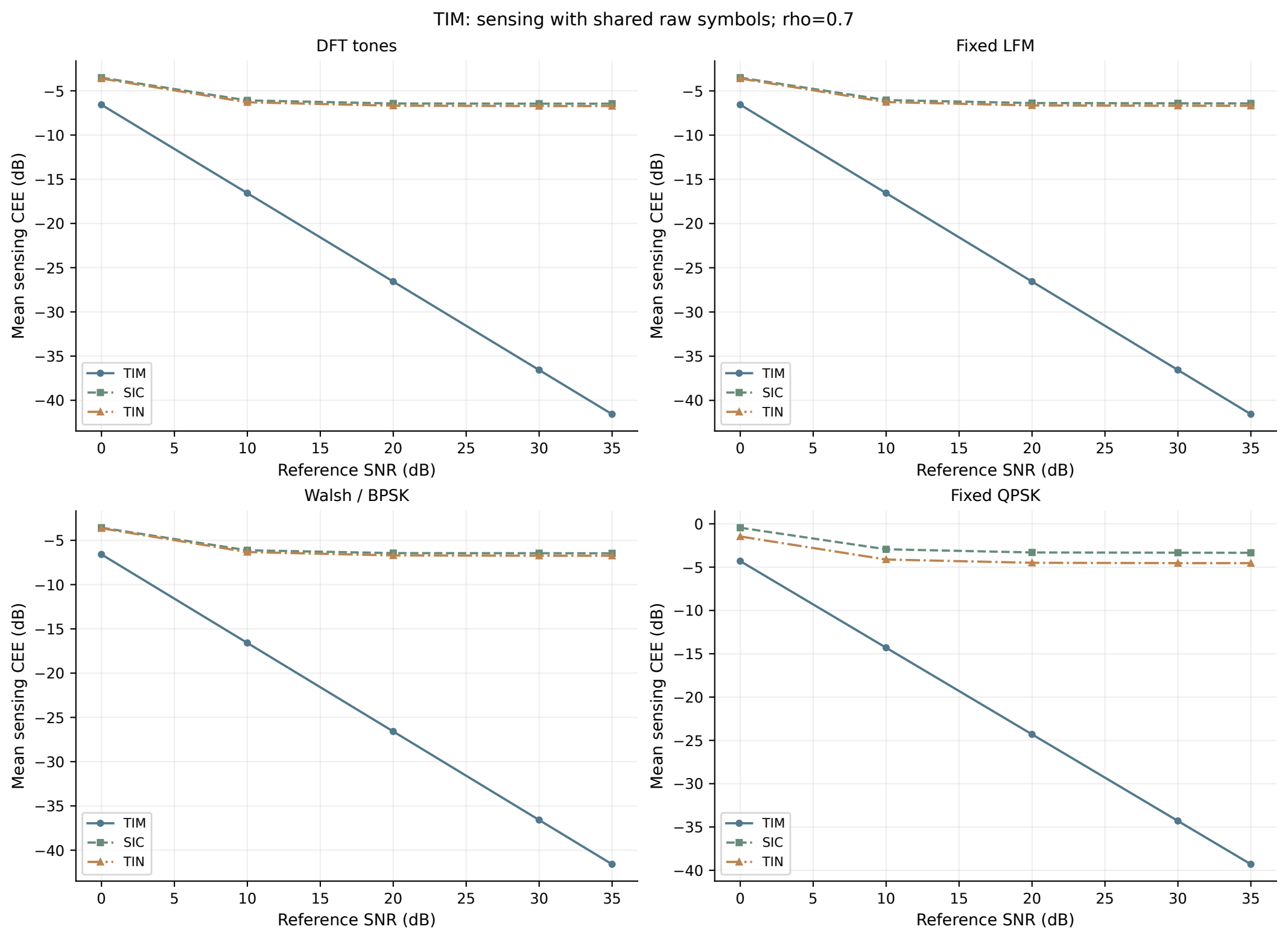

Fixed LFM: shared-source power tradeoffs, common window
BIA, SNR = 10 dB
BIA, SNR = 30 dB
TIM, SNR = 10 dB
TIM, SNR = 30 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, common window
(bit/channel use)

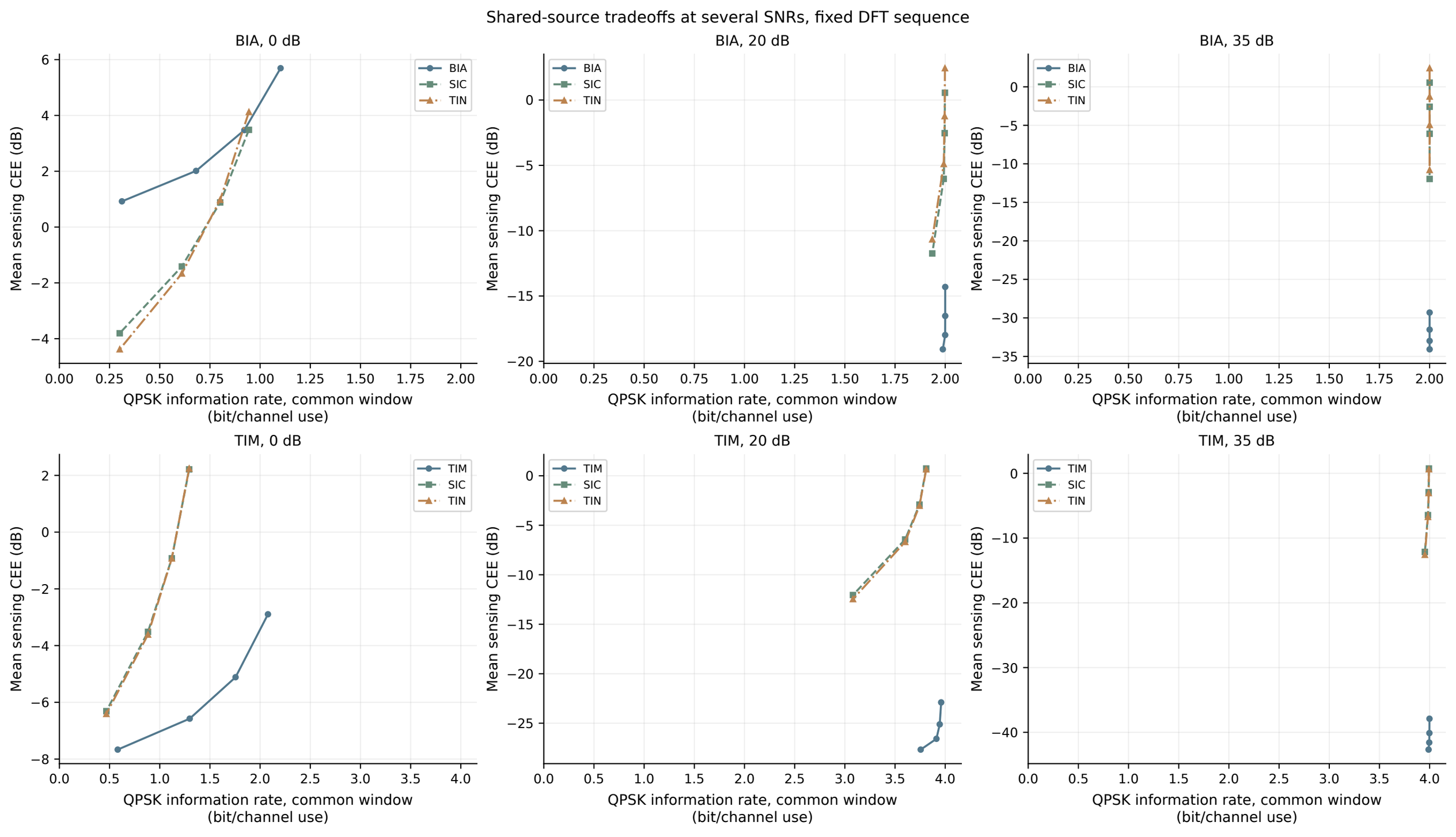

Walsh / BPSK: shared-source power tradeoffs, common window
BIA, SNR = 10 dB
BIA, SNR = 30 dB
TIM, SNR = 10 dB
TIM, SNR = 30 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, common window
(bit/channel use)

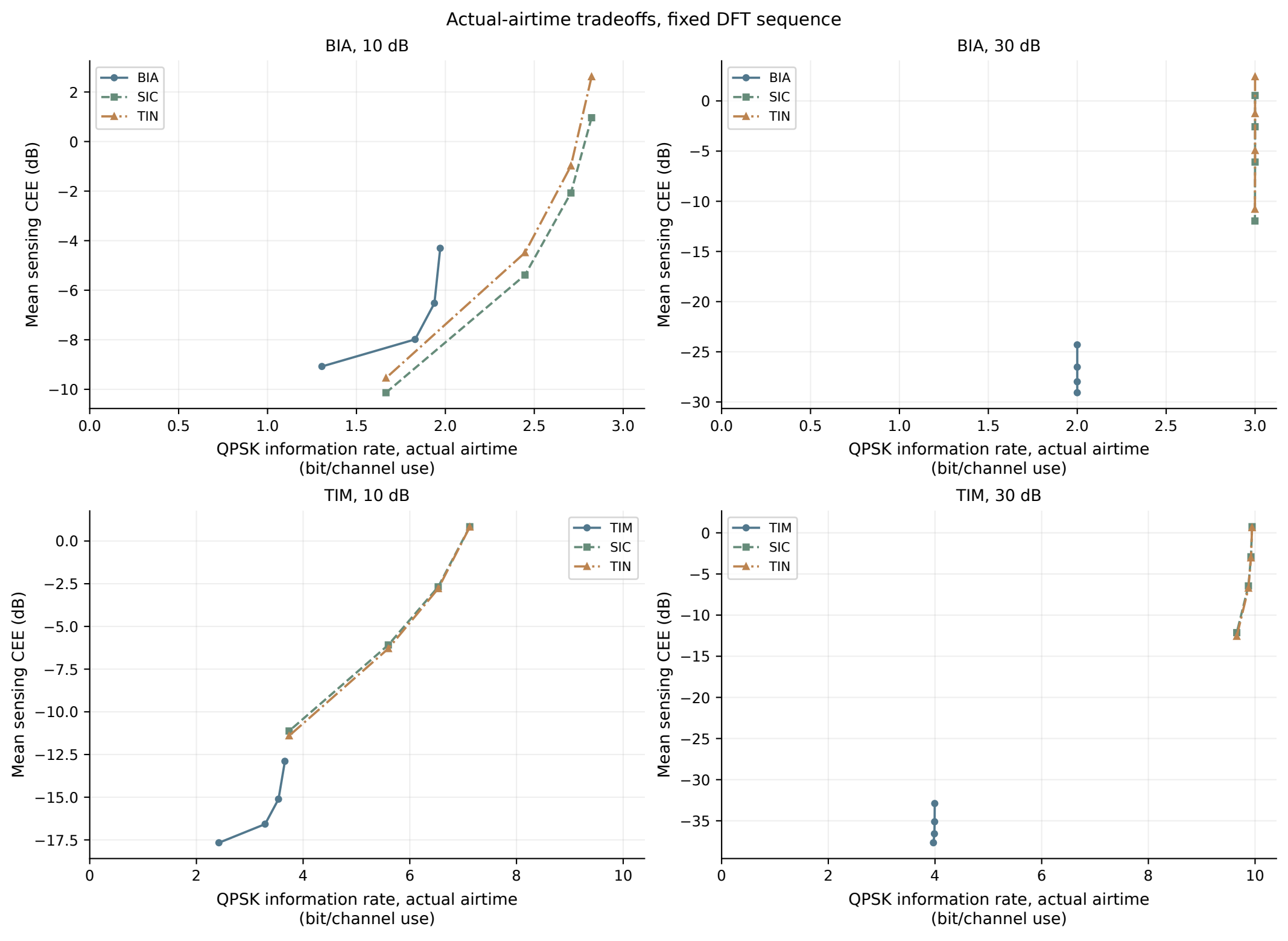

Fixed QPSK: shared-source power tradeoffs, common window
BIA, SNR = 10 dB
BIA, SNR = 30 dB
TIM, SNR = 10 dB
TIM, SNR = 30 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, common window
(bit/channel use)


Fixed sensing sequences under original encoding; rho=0.7
BIA
TIM
DFT tones
Fixed LFM
Walsh / BPSK
Fixed QPSK
Mean sensing CEE (dB)
Reference SNR (dB)
5
0
−5
−10
−15
−20
−25
−30
−35
−40
0
5
10
15
20
25
30
35


BIA: sensing with shared raw symbols; rho=0.7
DFT tones
Fixed LFM
Walsh / BPSK
Fixed QPSK
BIA
SIC
TIN
Mean sensing CEE (dB)
Reference SNR (dB)
0
−5
−10
−15
−20
−25
−30
−35
5
0
5
10
15
20
25
30
35


TIM: sensing with shared raw symbols; rho=0.7
DFT tones
Fixed LFM
Walsh / BPSK
Fixed QPSK
Mean sensing CEE (dB)
Reference SNR (dB)
TIM
SIC
TIN
0
5
10
15
20
25
30
35
−5
−10
−15
−20
−25
−30
−35
−40


Shared-source tradeoffs at several SNRs, fixed DFT sequence
BIA, 0 dB
BIA, 20 dB
BIA, 35 dB
TIM, 0 dB
TIM, 20 dB
TIM, 35 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, common window
(bit/channel use)


Actual-airtime tradeoffs, fixed DFT sequence
BIA, 10 dB
BIA, 30 dB
TIM, 10 dB
TIM, 30 dB
BIA
SIC
TIN
TIM
Mean sensing CEE (dB)
QPSK information rate, actual airtime
(bit/channel use)